Graduate Algorithms

Tutorial 1: Math for Algorithms

August 10, 2026

Why Study Algorithms?

Two undeniably true things about the human experience:

1. We are absolutely surrounded by problems
2. Our time is far too limited to spend forever solving them

This is the entire motivation for the study of algorithms.

Long before computers existed, people developed algorithms for arithmetic, navigation, voting, logistics, and trade.
Computers simply gave us the ability to apply those ideas at impossible scales.

Key Questions in Algorithms

In this course, we'll study how to think about problems efficiently.

- How do we compare different solutions?
- What makes one algorithm faster than another?
- How do we model complicated real-world situations?
- When is a problem easy, and when is it fundamentally difficult?
- How do we deal with 'difficult' problems?

The world is full of problems. Might as well get good at solving them.

Asymptotic Dominance

Definition

Asymptotic Dominance is a relation over functions, where $f(n) \prec g(n)$ when $g(n)$ approaches infinity faster than $f(n)$:

$$f(n) \prec g(n) \iff \limsup_{n\to\infty} \frac{|f(n)|}{|g(n)|} = 0$$

Furthermore, $f(n) \asymp g(n) \iff \limsup_{n\to\infty} \frac{|f(n)|}{|g(n)|} = k$ where $0 < k < \infty$ is constant.

Properties of Asymptotic Dominance

Corollary

Asymptotic dominance is transitive.

Corollary

If $\alpha < \beta$ *then* $x^\alpha \prec x^\beta$

Corollary

Let $f(x) = k \cdot g(x)$ *for some real* $k \neq 0$, *then* $f(x) \asymp g(x)$

Key insight: In a sum of functions, we only need to consider the most dominant function. We can talk about groups of functions using a single 'representative' function.

Representative Function

Definition

The representative function of a set of asymptotically equivalent functions is:

- A function with a leading coefficient of 1
- No additive subdominant terms

For example: x^2 is the representative function of the equivalence class of quadratic functions $ax^2 + bx + c$.

Theorem: Representative Functions

Theorem

If $f(x), g(x)$ are representative functions of equivalence classes F, G; then

$$f(x) \succ g(x) \iff \hat{f}(x) \succ \hat{g}(x)$$

for all $\hat{f} \in F$ and $\hat{g} \in G$

This means the asymptotic ordering is **invariant under scaling and translation by dominated functions**.

Exercise: Hierarchy of Common Functions

Prove that:

$$1 \prec \log\log n \prec \log n \prec n^{\epsilon} \prec n^{c} \prec n^{\log n} \prec c^{n} \prec n! \prec n^{n} \prec c^{c^{n}}$$

where ϵ, c are small positive constants.

Big Oh Notation (Definition 1)

Definition (Big Oh (from hierarchy))

$$f(n) = O(g(n)) \iff f(n) \preceq g(n)$$

Definition (Big Oh (limit))

$$f(n) = O(g(n)) \iff \limsup_{n\to\infty} \frac{|f(n)|}{|g(n)|} < \infty$$

Definition (Big Oh (Classical))

$$f(n) = O(g(n)) \iff \exists c, n_0 \text{ s.t. } \forall n \geq n_0, |f(n)| \leq c \cdot |g(n)|$$

Bachmann-Landau Notation

Notation	Name
$f(n) = \Theta(g(n))$	Big Theta
$f(n) = \Omega(g(n))$	Big Omega
$f(n) \sim g(n)$	Total Asymptotic Equivalence
$f(n) = o(g(n))$	Small Oh
$f(n) = \omega(g(n))$	Small Omega

Example: $3n^2 - 100n + 6$

- $3n^2 - 100n + 6 = O(n^2)$
- $3n^2 - 100n + 6 = O(n^3)$
- $3n^2 - 100n + 6 \neq O(n)$
- $3n^2 - 100n + 6 = \Omega(n^2)$
- $3n^2 - 100n + 6 \neq \Omega(n^3)$
- $3n^2 - 100n + 6 = \Theta(n^2)$
- $3n^2 - 100n + 6 \neq \Theta(n^3)$

Also: $3n^2 - 100n + 6 \sim 3n^2$ by taking the limit.

Exercise: Big Oh Arithmetic

Prove the following:

1. $n^m = O(n^{m'})$ for $m \leq m'$
2. $O(f(n)) + O(g(n)) = O(|f(n)| + |g(n)|)$
3. $f(n) = O(f(n))$
4. $O(O(f(n))) = O(f(n))$
5. $c \cdot O(f(n)) = O(f(n))$ for constant c
6. $O(f(n)) \cdot O(g(n)) = O(f(n)g(n))$
7. $O(f(n)^2) = O(f(n))^2$

Sum of Powers: Asymptotic Behavior

Example

Consider $f_k(n) = \sum_{i=1}^{n} i^k$ for $k, n \in \mathbb{N}$.
Small cases:

$$f_0(n) = \sum_{i=1}^{n} 1 = n = O(n)$$

$$f_1(n) = \sum_{i=1}^{n} i = \frac{n(n+1)}{2} = O(n^2)$$

$$f_2(n) = \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} = O(n^3)$$

Theorem: Sum of Powers

Theorem

For all $k \geq 0$*:*

$$f_k(n) = \sum_{i=1}^{n} i^k = O(n^{k+1})$$

Proof: Sum of Powers

Proof.

$$
\begin{aligned}
f_k(n) &= \sum_{i=1}^{n} i^k \\
&= \sum_{i=1}^{n} O(n^k) \\
&= n \cdot O(n^k) \\
&= O(n^{k+1})
\end{aligned}
$$

$\square$

This proof is simpler than induction!

Why Graphs?

Statements we often hear:

- "Your friends, on average, have more friends than you"
- "Any 2 people can be connected to each other in 6 relations"

One way to reason about networks (people, roadways, etc.) is through **Graphs**.

We'll formalize and prove such claims using graph theory.

Simple Graph

Definition

A (simple) graph is an ordered pair $G = (V, E)$ where:

- V is the set of **vertices**
- $E \subseteq \{\{x, y\} \mid x, y \in V, x \neq y\}$ is a set of unordered pairs called **edges**

Visualization: Each vertex is a point in the plane; each edge is a line between corresponding vertices.

Directed Graph

Definition

In a directed graph, $E \subseteq \{(x, y) \mid x, y \in V, x \neq y\}$. The pairs are ordered, making the edges **directed**.

Visualization: Edges are drawn with arrowheads to indicate direction.

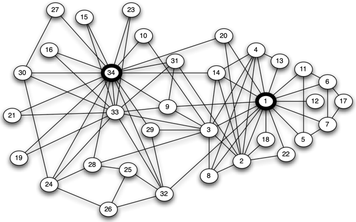
Weighted (Directed) Graph

Definition

A (directed) graph with a function $w : E \to \mathbb{R}$ is called a weighted (directed) graph.

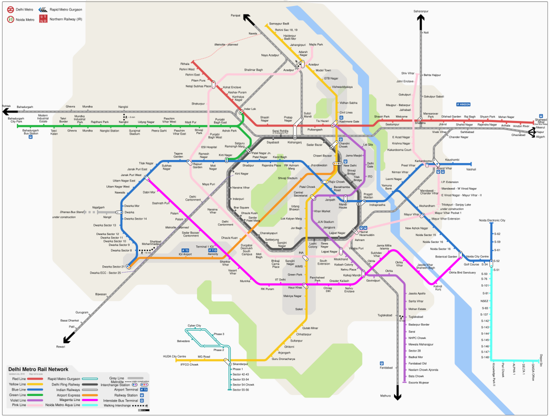
This allows modeling real-world quantities like distances, costs, or capacities.

Example: Karate Club Network



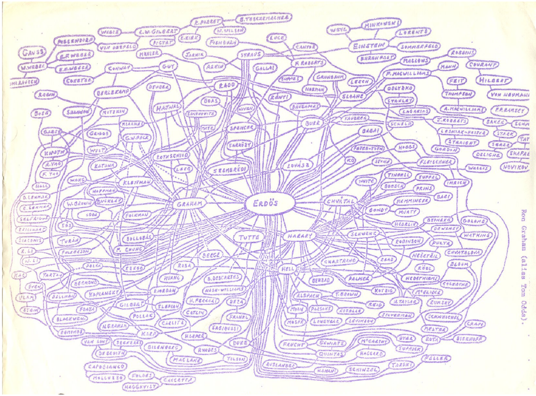
The social network of friendships within a 34-person karate club.
Questions: If we wanted to spread a rumor quickly, where would we start? How to split the club?

Example: Delhi Metro



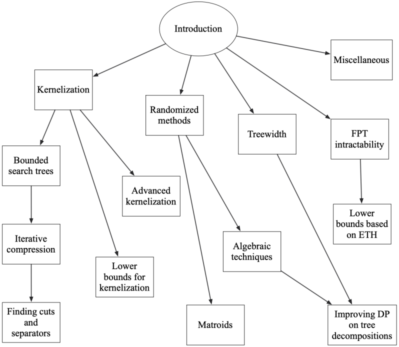
Question: How would we figure out a route between two locations? Can we draw it more legibly?

Example: Erdős Collaboration Network



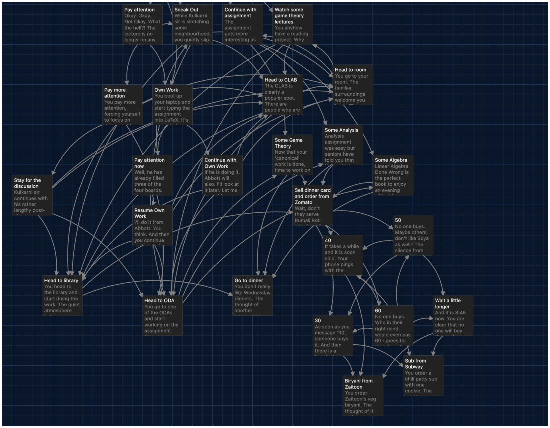
Handdrawn graph of Erdős's collaborations.
Question: Given a person, can we find their Erdős number efficiently?

Example: Chapter Dependencies



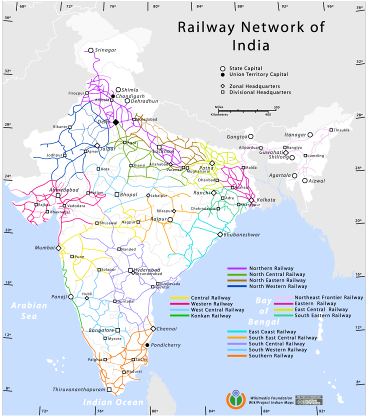
Chapterwise dependencies from Parametrized Algorithms.
Questions: In what order should chapters be read? Which is most important?

Example: Game Choice Tree



Part of a choice tree for a game.
Questions: Is there a loop? Which scenes are visited most?

Example: Indian Rail Network



Rail network of India.

Questions: What is the cheapest/fastest route? Which rails should we shut down to cut costs while maintaining connectivity?

Core Insight

There are many graphs and many questions about them:

- Shortest paths
- Network connectivity
- Centrality and influence
- Cycles and trees
- Optimal subgraphs

This is what we'll explore in coming classes and tutorials.

Subgraph

Definition

We say $H = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

Informally: H is obtained by removing vertices and edges from G.

Adding/Removing Vertices & Edges

Definition

For an edge xy or vertex x:

- $G - xy$ is G with edge xy removed
- $G - x$ is G with vertex x and all its incident edges removed
- $G + xy$ is G with edge xy added
- $G + x$ is G with vertex x added

Neighborhood and Degree

Definition

If $xy \in E$, then x and y are **adjacent** or **neighbors**.
The **neighborhood** of x is: $N(x) = \{y \in V \mid xy \in E\}$

Definition

The **degree** of a vertex x is: $\deg(x) = |N(x)|$
This equals the number of edges incident to x.

Handshake Lemma

Lemma

For all graphs $G = (V, E)$:

$$\sum_{v \in V} \deg(v) = 2|E|$$

Proof.

Every edge contributes to the degrees of exactly 2 vertices. Thus, adding the degrees counts each edge exactly twice. $\square$

Your Friends Have More Friends

Theorem

For a graph $G = (V, E)$ with no isolated vertices:

$$\sum_{v \in V} \deg(v) \leq \sum_{v \in V} \frac{\sum_{n \in N(v)} \deg(n)}{\deg(v)}$$

The average degree of neighbors exceeds the average degree of vertices.

Proof: Your Friends Have More Friends

Proof.

By the Handshake Lemma, LHS $= 2|E|$.
For the RHS:

$$\begin{aligned}
\sum_{v \in V} \frac{\sum_{n \in N(v)} \deg(n)}{\deg(v)} &= \sum_{v \in V} \frac{\sum_{uv \in E} \deg(u)}{\deg(v)} \\
&= \sum_{uv \in E} \left(\frac{\deg(u)}{\deg(v)} + \frac{\deg(v)}{\deg(u)} \right) \\
&\geq 2|E|
\end{aligned}$$

The last inequality uses: $\frac{a}{b} + \frac{b}{a} \geq 2$ for all positive a, b. $\square$

Implications (1): The Comparison Trap

Comparison is truly the thief of joy.

If you look at:

- The number of friends of your friends
- Citations of people you cite
- Followers of people you follow
- Partners of your partner

You will likely feel sad. The graph gives you good reason to believe everyone else is doing better, but this tells you nothing about how well you're actually doing.

(In many cases, the graph is weighted, and you might not be accounting for the weights!)

Implications (2): Finding High-Centrality Vertices

To find a high-centrality individual:
Method: Randomly choose a vertex and randomly choose one of its neighbors.

Why this works:

- The high-degree vertices will be neighbors of many vertices
- So they're more likely to be selected

Even better: Choose a vertex and pick its neighbor with highest degree.
Applications: Preventing rumors, controlling pandemics.

Example: Finding Influential People

Instagram Example:
We can find the most followed person by asking a random person who the most famous person they follow is.

- Probability of success: $\approx \frac{\text{678 million}}{\text{3 billion}} \approx 0.226$
- Much better than random guessing!

Time Complexity:

- Computing all degrees: $O(|V|)$
- Random approach: $\mathbb{E}[O(\deg(v))] = O\left(\frac{2|E|}{|V|}\right)$